Aim:

(i) To perform transient analysis and AC response on series resonance and parallel resonance RLC circuit.

(ii) Plot bode Plot and match theoretical result with practical result.

Theory:

Resonance is defined as natural frequency of the system concide with driving force frequency. Its application is used in communication system and filter design.

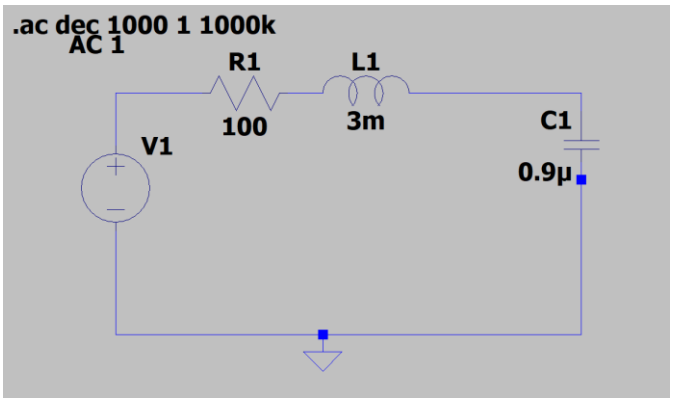
Electrical Resonance means in a circuit when the inductive reactance (XL) and capacitive reactance (XC) are equal in magnitude but opposite in phase, resulting in a purely resistive impedance at a particular frequency. This condition leads to several distinctive effects and applications in AC circuits. resonance occurs when the energy transfer from an external vibratory force to a system is maximized due to matching frequencies, resulting in large amplitude oscillations.

Resonance in an electrical circuit occurs when the circuit's inductive reactance (XL) and capacitive reactance (XC) are equal in magnitude but opposite in phase, resulting in a purely resistive impedance at a particular frequency. This condition leads to several distinctive effects and applications in AC circuits

History about Resonance:

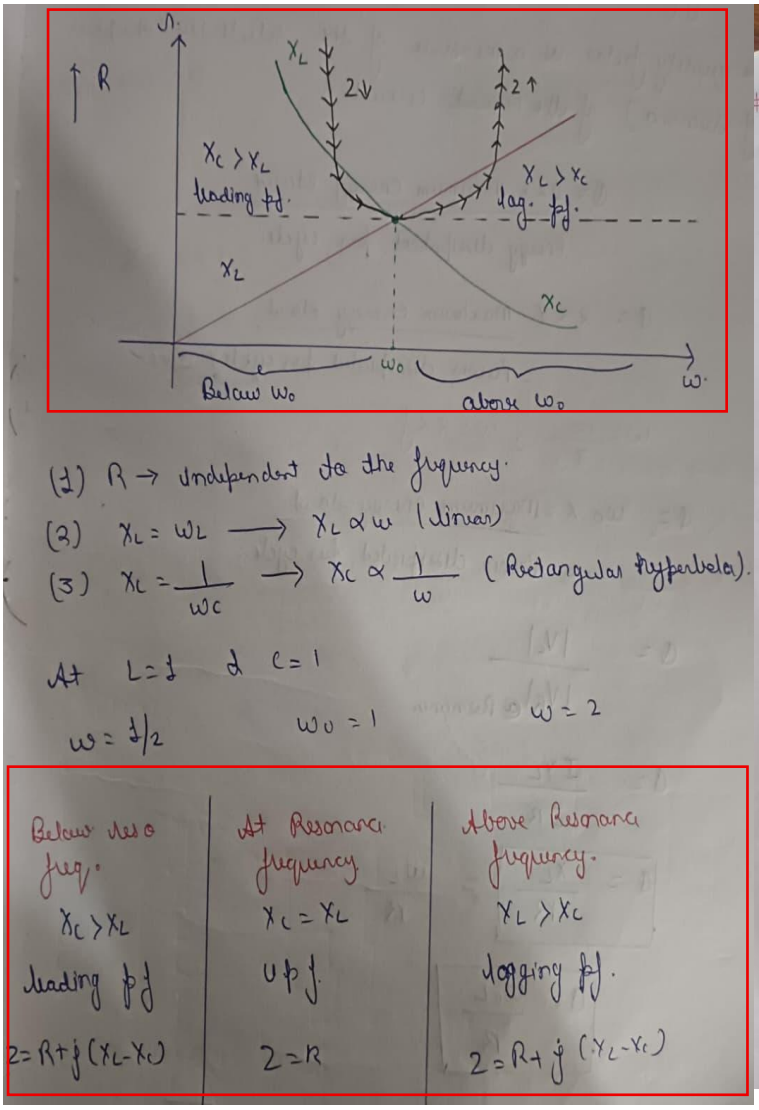
Actually, resonance was the phenomena which came into picture when suspension type bridge got collapsed during a march past by an army while passing through the bridge. The reason of collapse after investigation was found that the natural frequency of bridge was matched with army's unit march past frequency. This phenomena of natural frequency for electrical become blessing in disguise as by using these phenomena many communication systems, in power electronics and power system as harmonic filters for inverter-based circuit's. In transmission line this concept is often used impedance matching while performing the calculation of long transmission line.

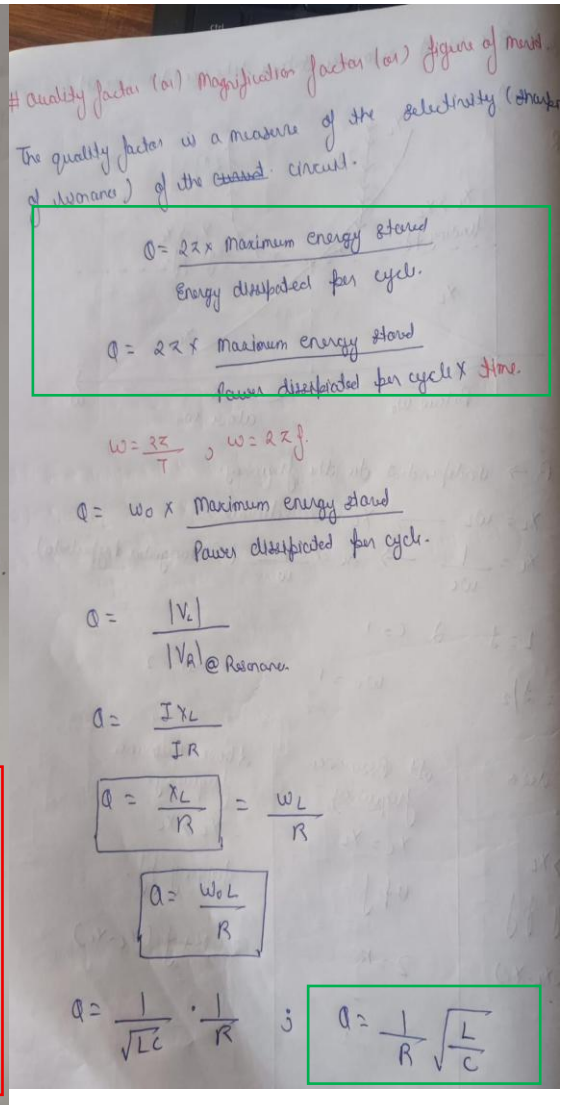
Series Resonance:



Series resonance circuit behaves like and acceptor circuit. This condition occurs when inductive reactance is equal to capacitive reactance, at this condition the impedance is minimum and current

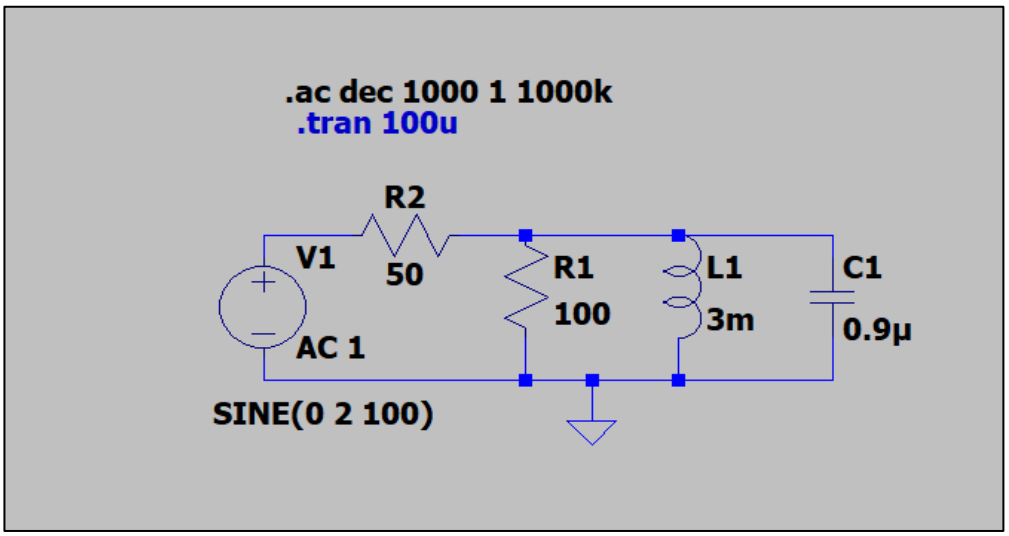
and voltage are in same phase, and current has the maximum value.This circuit is also called voltage magnification circuit.



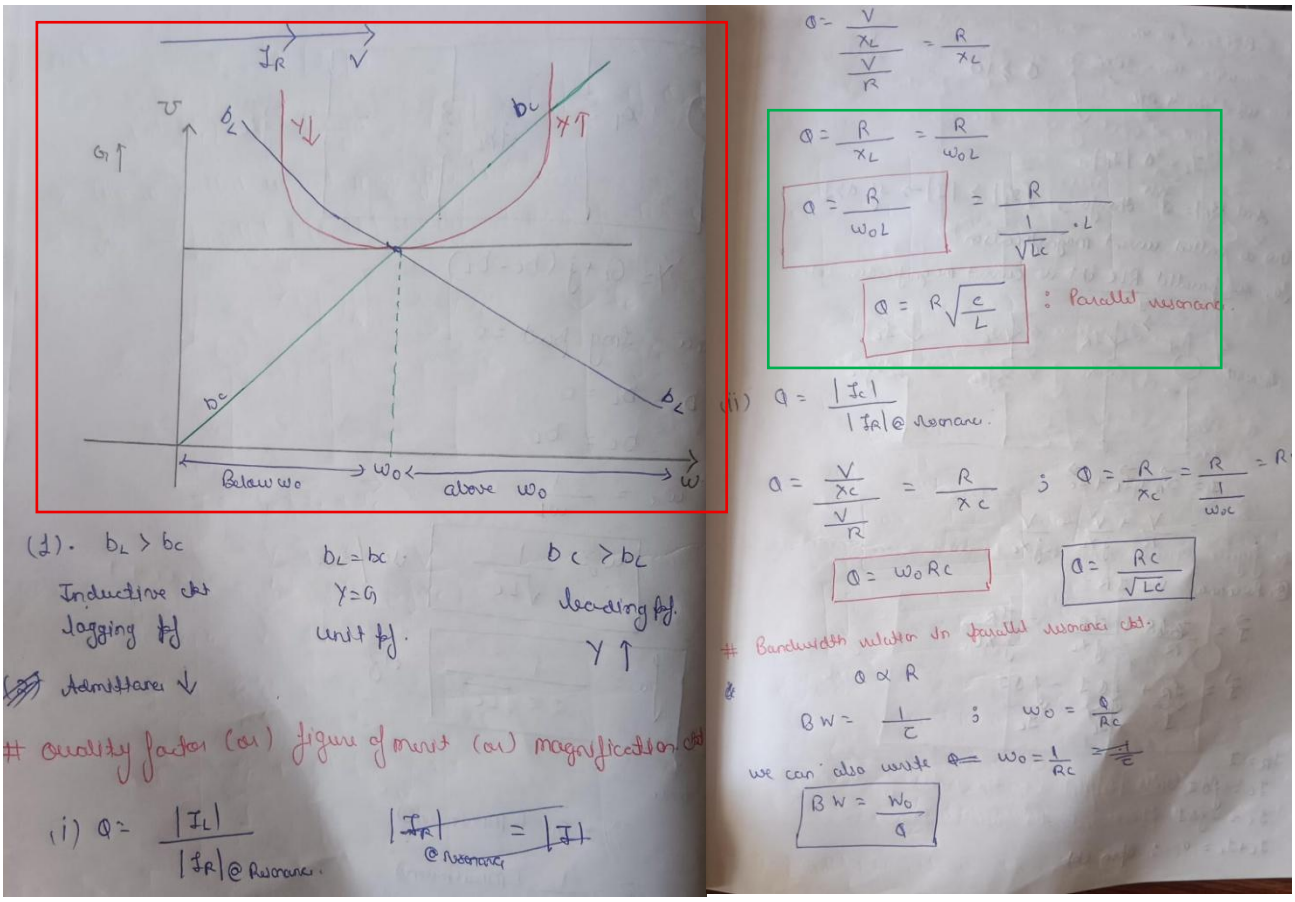


Graph for impedance v/s frequency having different values of inductive reactance XL and Capacitive reactance XC.At Wo XL=XC which occurs at resonance frequency and about the quality factor of the circuit

Parallel Resonance:

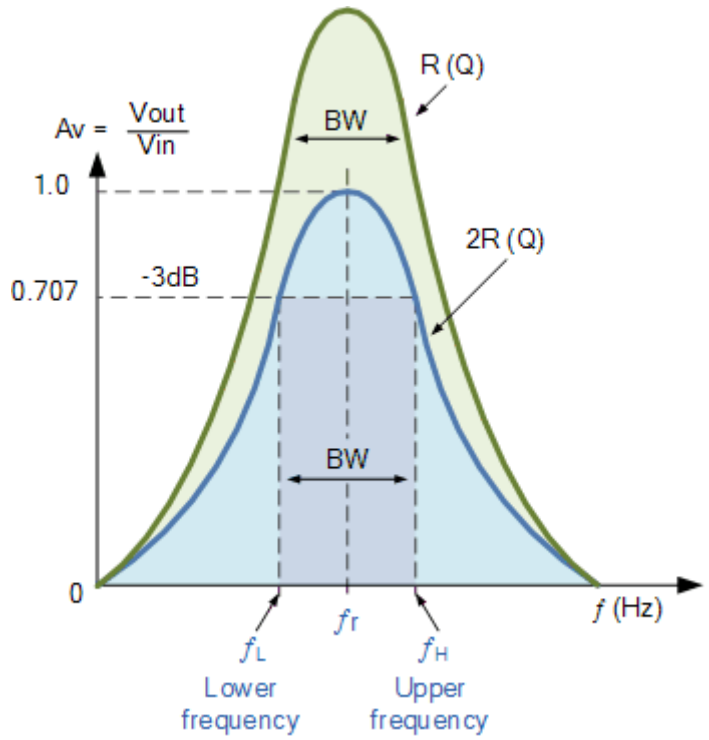


In a parallel resonance circuit, the inductor (L) and capacitor (C) are connected in parallel, with a resistor (R) typically in series with the inductor. At the resonant frequency (f_o), the impedance of the inductor and capacitor cancel each other out, resulting in a sharp increase in current flow through the circuit.



Graph for admittance v/s frequency having different values of inductive reactance bL and Capacitive reactance C.At Wo(resonant frequency) bL=bC which occurs at resonance frequency and about the quality factor of the circuit

Selectivity of the circuit:



Selectivity of the circuit depends on the measure on the quality factor of the circuit. The higher the circuit quality factor the smaller will be the bandwidth.

B.W.=fo/Q;

Where fo: Resonant frequency

Q:Quality factor.

Bode Plot:

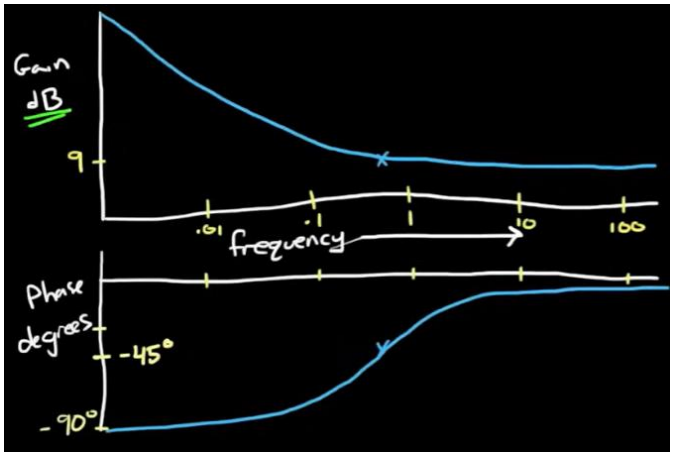
Before discussing about bode plot we must discuss about the frequency domain analysis.

In general there are sinusoidal signals in communication systems. These sinusoidal signals is very tedious task to analyse in time domain. So, to reduce this task we do analysis of sinusoidal signal in frequency domain.

In frequency domain analysis because there is change in amplitude and phase but not in frequency.

Bode Plot: In bode plot we need magnitude and phase where magnitude is plot in (dB) and phase is plot (degree) against w (omega) but in logarithmic scale.

Bode plot will give result for the different frequency spectrum by plotting on e for phase and one for gain.



Bode plot for frequency and phase

Source: Brian Doughlas

Procedure to draw Bode plot.

① $G(s)\,H(s) = \text{gain} \rightarrow \dfrac{K(1+s)}{s^{\cancel{2}}(1+2s)}$ — ①

zero

finite poles

Pole @ origin

② put $s = j\omega$ (in eq ①)

③ $G(j\omega)\,H(j\omega) = \dfrac{K(j+\omega)}{j\omega(1+2j\omega)}$

④ Find magnitude and convert into dB

$$M = \frac{K\sqrt{1+\omega^2}}{\omega\sqrt{1+4\omega^2}}$$

$$M_{dB} = \log_{10}\left[\frac{K\sqrt{1+\omega^2}}{\omega\sqrt{1+4\omega^2}}\right]$$

$$M_{dB} = \underbrace{\log_{10}(K\sqrt{1+\omega^2})}_{m,\ n} - \underbrace{\log_{10}(\omega\sqrt{1+4\omega^2})}_{n}$$

$$M_{dB} = \log_{10} K + \log_{10}\sqrt{1+\omega^2} - \log_{10}\omega + \log_{10}\sqrt{1+4\omega}$$

⑤ Find the phase.

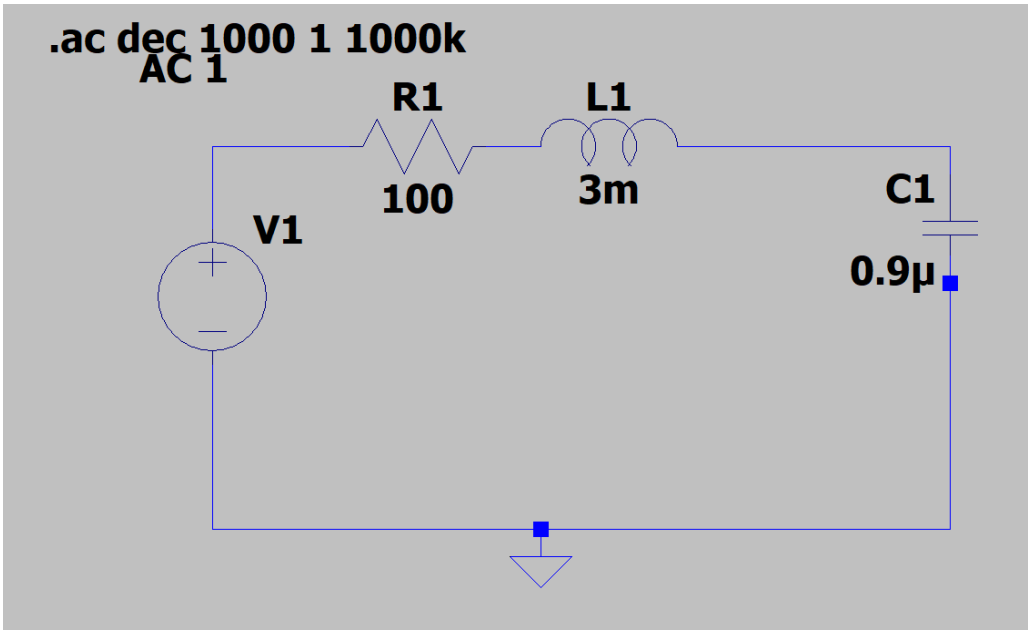
$$G(s)\,H(s) = \frac{K(1+s)}{s(1+2s)}$$

$$\phi = 0 + \tan^{-1}\omega - 90 - \tan^{-1}2\omega.$$

⑥ Plot the values of mag. & phase according to the

Simulation Results:

(i) Series Resonance



This is the AC analysis of the circuit:

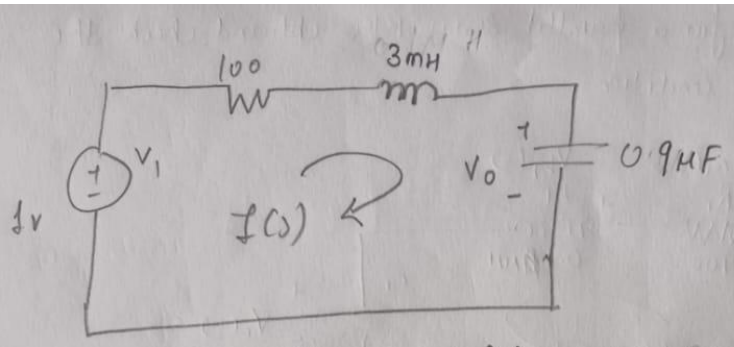
Hand calculation according bode plot procedure:

This is the second order transfer function whose equation is given by (general equation)

$$\boxed{\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\delta\omega_n s + \omega_n^2}}$$

This is the second order equation transfer function.

Series RLC circuit transfer function obtained is as follows:



$R_1 \to R_1$, $L_1 \to L_1(s)$ $C_1 \to \frac{1}{C(s)}$

$$\frac{V_1(s) - V_o(s)}{V_c(s)} = \frac{\cancel{I(s)}\left[R_0 + Ls + \frac{1}{C(s)}\right]}{\cancel{I(s)} \times \frac{1}{C_1(s)}} \quad (1)$$

$$= \frac{R + Ls + \frac{1}{C(s)}}{\frac{1}{C(s)}} = \frac{\frac{RCs + LCs + 1}{C(s)}}{\frac{1}{C(s)}}$$

$$\frac{V_o(s)}{V_c(s)} = \frac{1 + sCR + LCs^2}{\cancel{C(s)}} \times \cancel{C(s)}$$

$$\omega_0 = \frac{1}{\sqrt{0.1 \times 0.1}}$$

$$\frac{V_c(s)}{V_o(s)} = \frac{1}{1 + sCR + LCs^2}$$

$$TF = \frac{1}{1 + sCR + LCs^2} \quad \cancel{\Rightarrow \omega_n^2 = 1}$$

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{1 + 2\zeta\frac{s}{\omega_n} + \frac{s^2}{\omega_n^2}}$$

$$\boxed{TF = \frac{1/LC}{s^2 + (R/L)s + 1/LC}} \to \omega_n = \frac{1}{\sqrt{LC}}$$

$$2\zeta\omega_n = \frac{R}{L}$$

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

This the transfer function of the series RLC circuit which is converted in the form second order circuit and then it is being compared with the second order equation transfer function

$$\xi = \frac{R}{2}\sqrt{\frac{C}{L}} = \frac{100}{2}\sqrt{\frac{0.9\times10^{-6}}{3\times10^{-3}}} = 0.8666$$

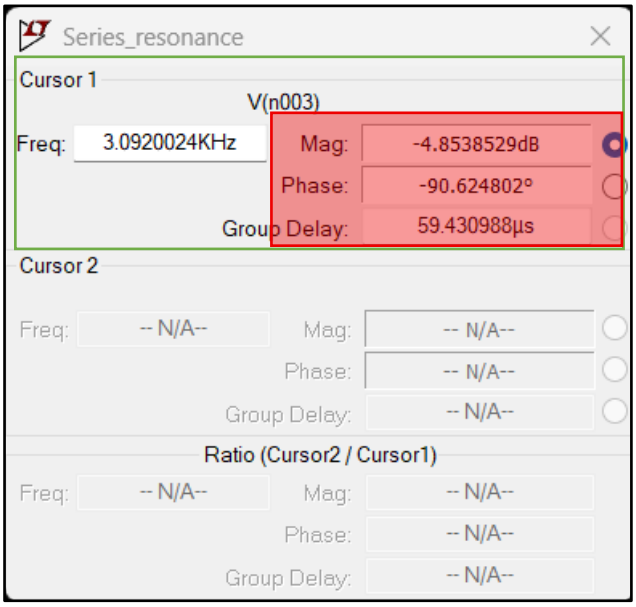
$$M_{dB} = -20\log 2\xi \quad \text{and} \quad \phi = -90^\circ \text{ or } -\pi/2$$

$$= -20\log 2\times0.8666$$

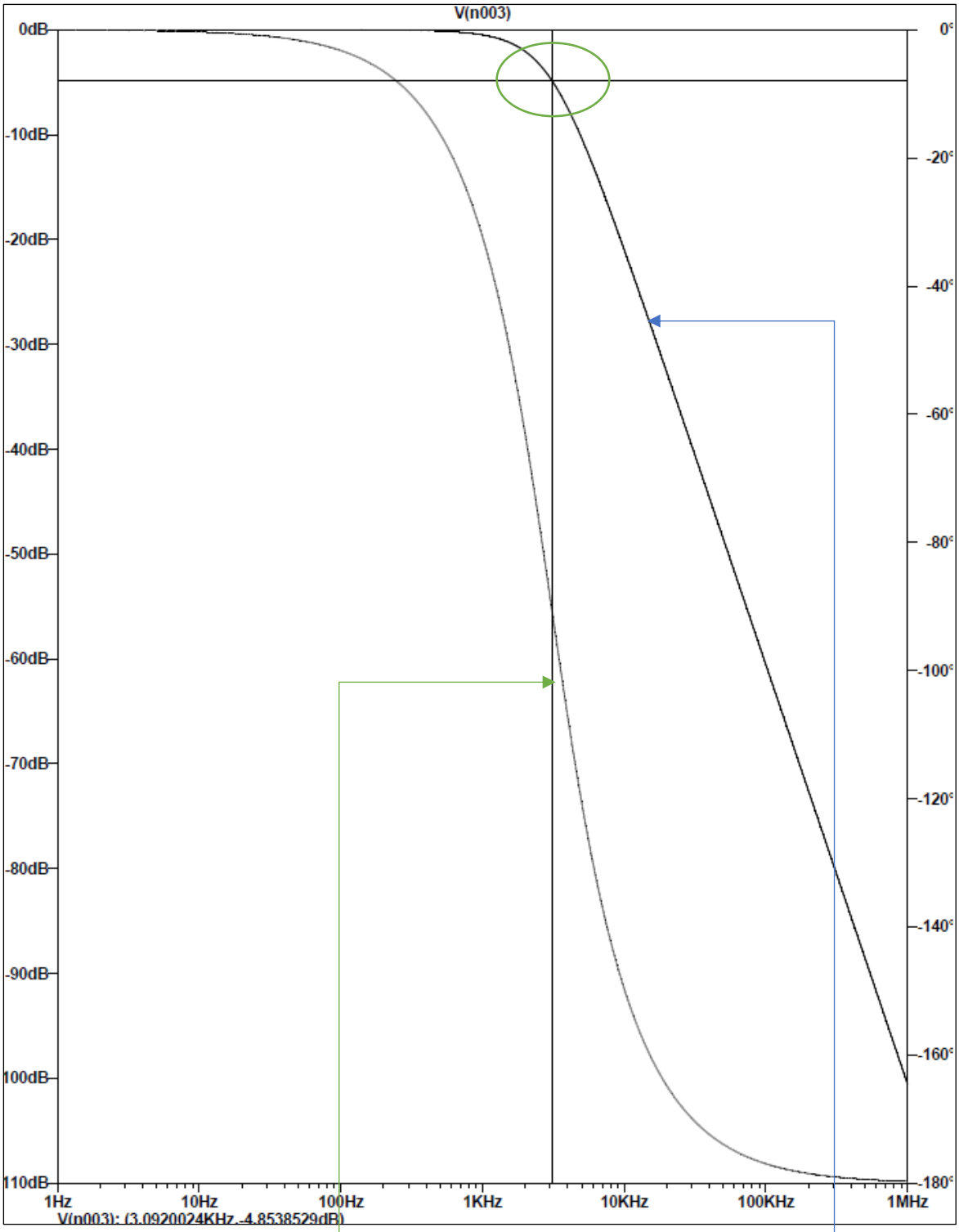
$$M_{dB} = -4.7709\,dB$$

$\xi \rightarrow$ Damping ratio

Practical Results:



We matched the value at **phase =-90 degree** and matched the value of magnitude at that particular phase.This is the point where the wo (circuit frequency) is equal to wn (natural frequency) i.e **w0/wn=1** and **Magnitude=-20log (2zeta).**



This the phase plot

This the magnitude plot

Magnitude and phase plot of the given series RLC circuit.

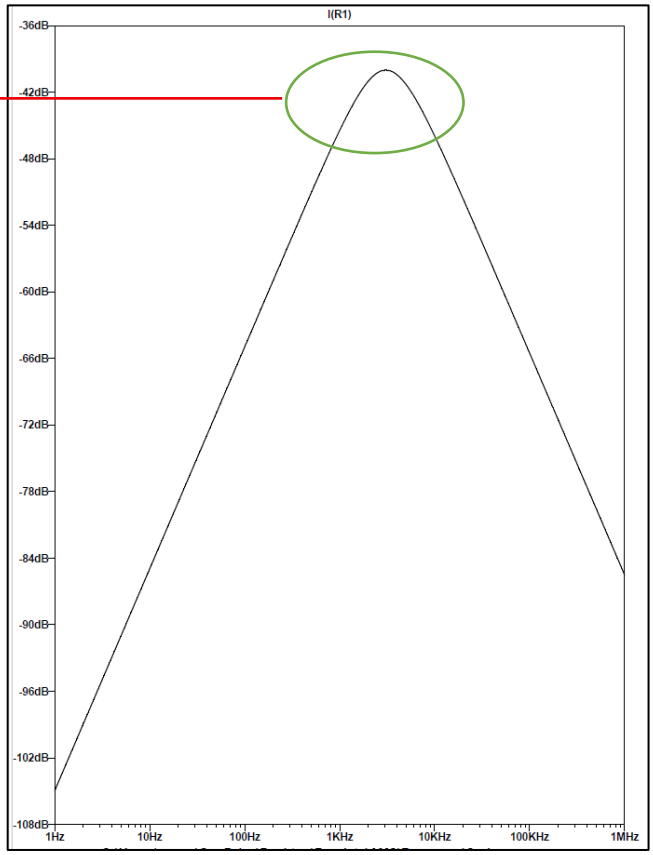
$$\omega_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow f_0$$

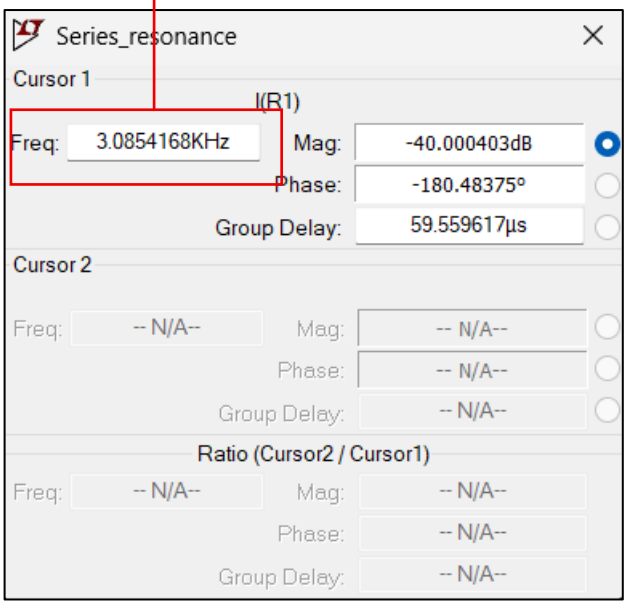
$$f_0 = \frac{1}{2\pi\sqrt{3\times10^{-3}\times0.9\times10^{-6}}} = 3062.94\ Hz$$

or

3.063 KHz.

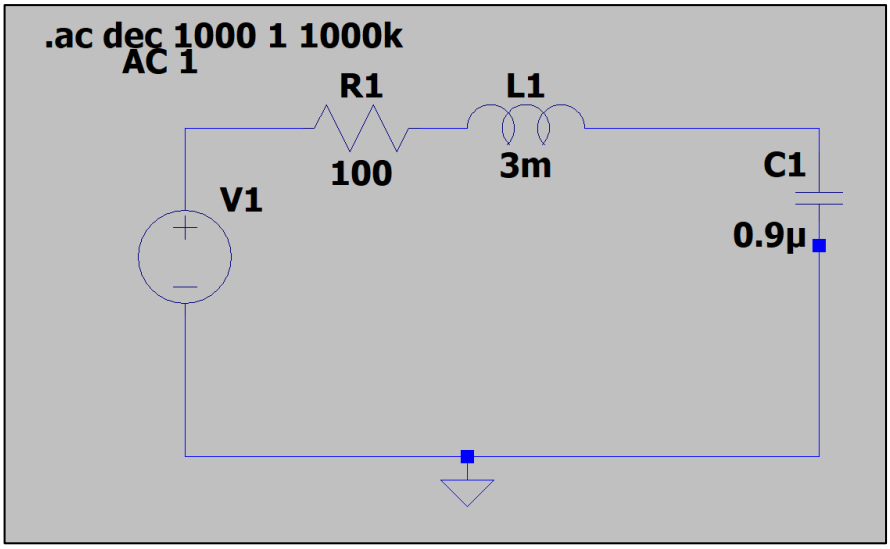
$$\boxed{f_0 = 3.063\ KHz.}$$

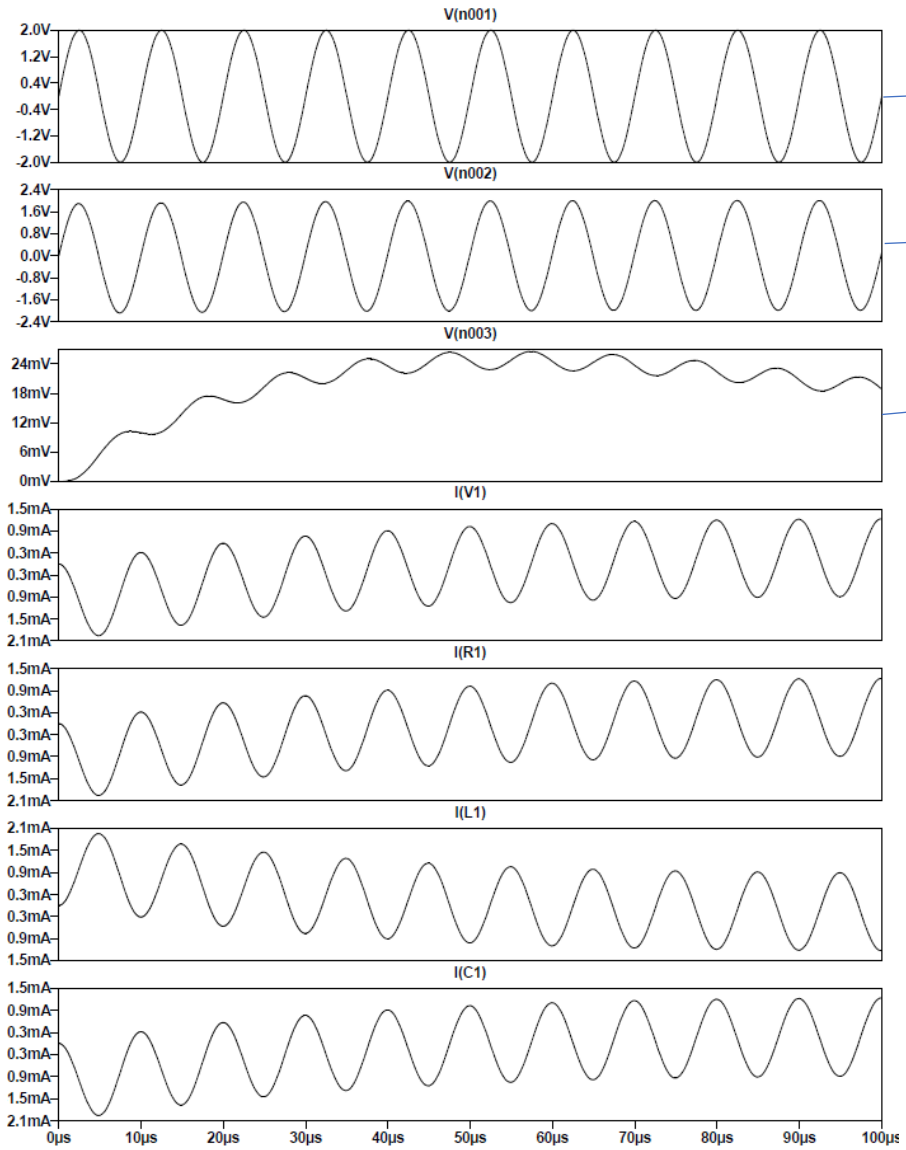




Resonance Frequency theoretical and resonance frequency simulated matched

Transient Analysis:



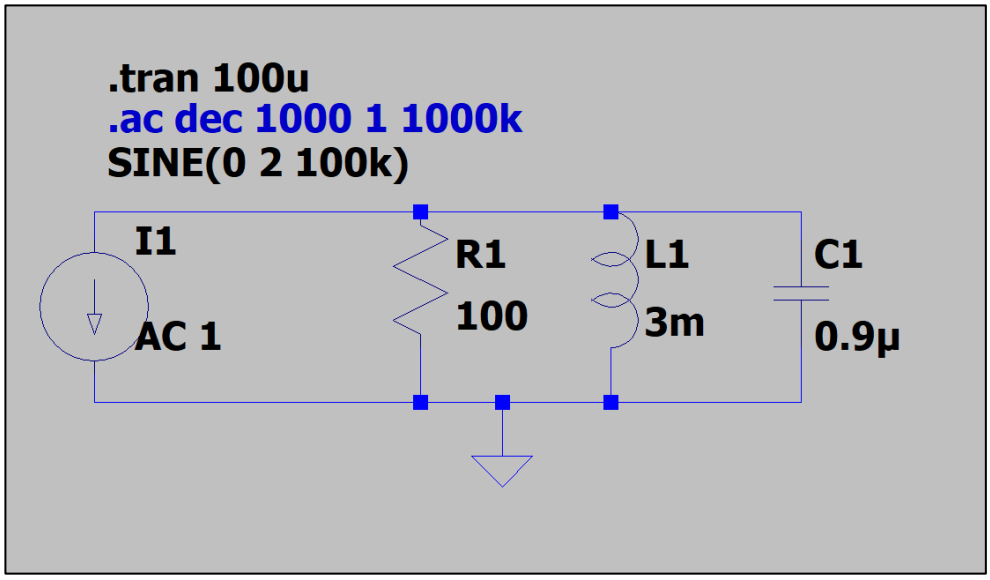


Voltage across Resistor

Voltage across Inductor

Voltage across Capacitor

Parallel Resonance:



This is the AC analysis of the circuit:

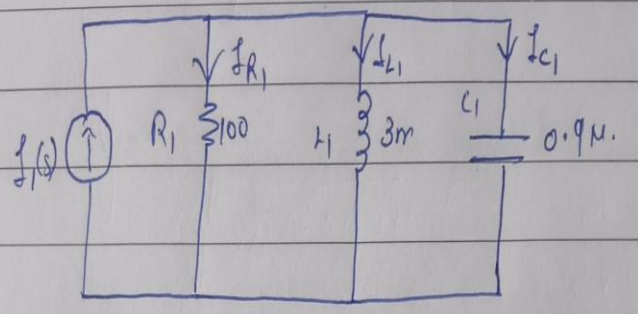
Hand calculation according bode plot procedure:

This is the second order transfer function whose equation is given by (general equation)

$$\boxed{\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\delta\omega_n s + \omega_n^2}}$$

This is the second order equation transfer function and below is the Transfer function calculation for Parallel Resonance circuit. The result obtained will be compared with the above equation,

Parallel resonance :-



$I_1 \rightarrow I_1(s)$, $I_{R_1} \rightarrow I_{R_1}(s)$, $I_{L_1} \rightarrow I_{L_1}(s)$

$I_{C_1} \rightarrow I_{C_1}(s)$

$R_1 \rightarrow R_1$, $L_1 \rightarrow L_1(s)$, $C_1 \rightarrow 1/C(s)$

$$I_1(s) = I_{R_1}(s) + I_L(s) + I_C(s)$$

$$I_1(s) = \frac{V(s)}{R_1} + \frac{V(s)}{sL} + V(s)\cdot sC$$

$$I_1(s) = V(s)\left[\frac{1}{R} + \frac{1}{sL} + sC\right]$$

$$\frac{\cancel{I_1(s)}\ V(s)}{I_1(s)} = \frac{1}{\frac{1}{R} + \frac{1}{sL} + sC}$$

$$H(s) = \frac{s/C}{s^2 + s/RC + 1/LC}$$

$Q = R\sqrt{\frac{C}{L}}$: In parallel resonance

$$\boxed{Q = 100\sqrt{\frac{0.9\times10^{-6}}{3\times10^{-3}}} = 1.73205}$$

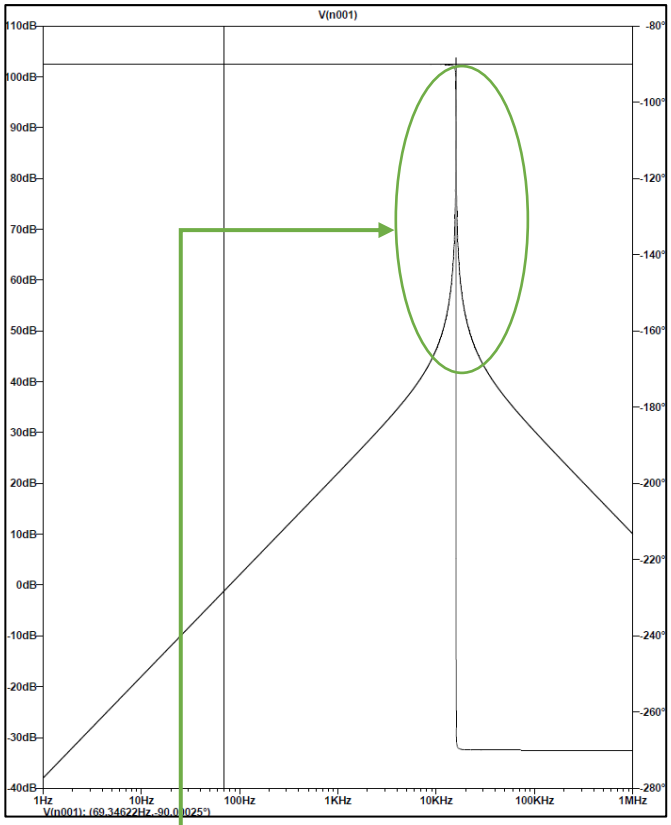
$\cancel{2\xi}$ $2\xi = 1/Q$; $\xi \rightarrow$ Damping factor.

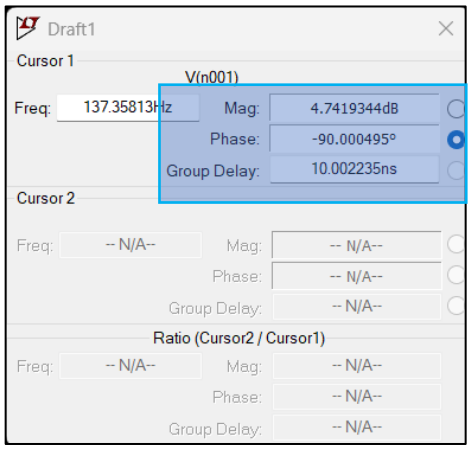
$$M = -20\log 2\xi = -20\log 0.57735$$

$$\boxed{M_{dB} = 4.77121}$$

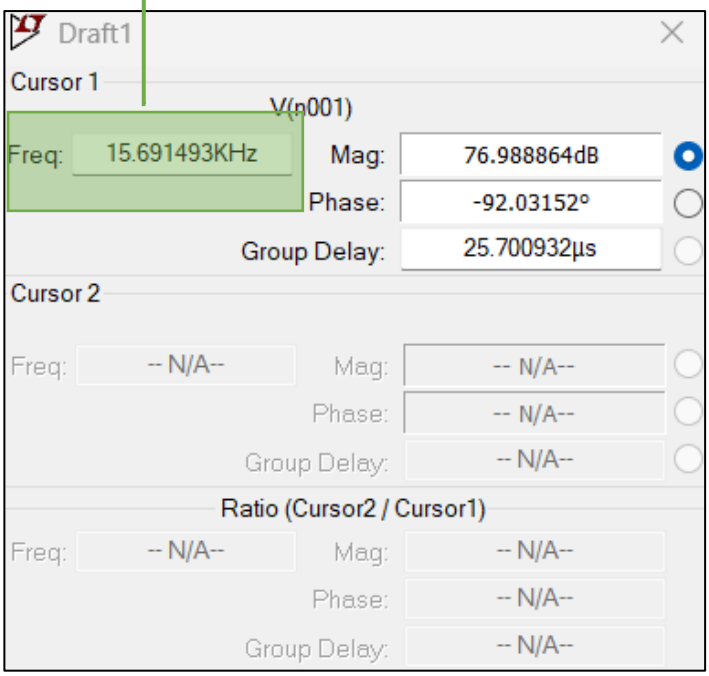
$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{3\times10^{-3}\times0.9\times10^{-6}}} = \frac{1}{\sqrt{2.7\times10^{-9}}}$$

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{2.7\times10^{-9}}} = 3.06294\ KHz.}$$



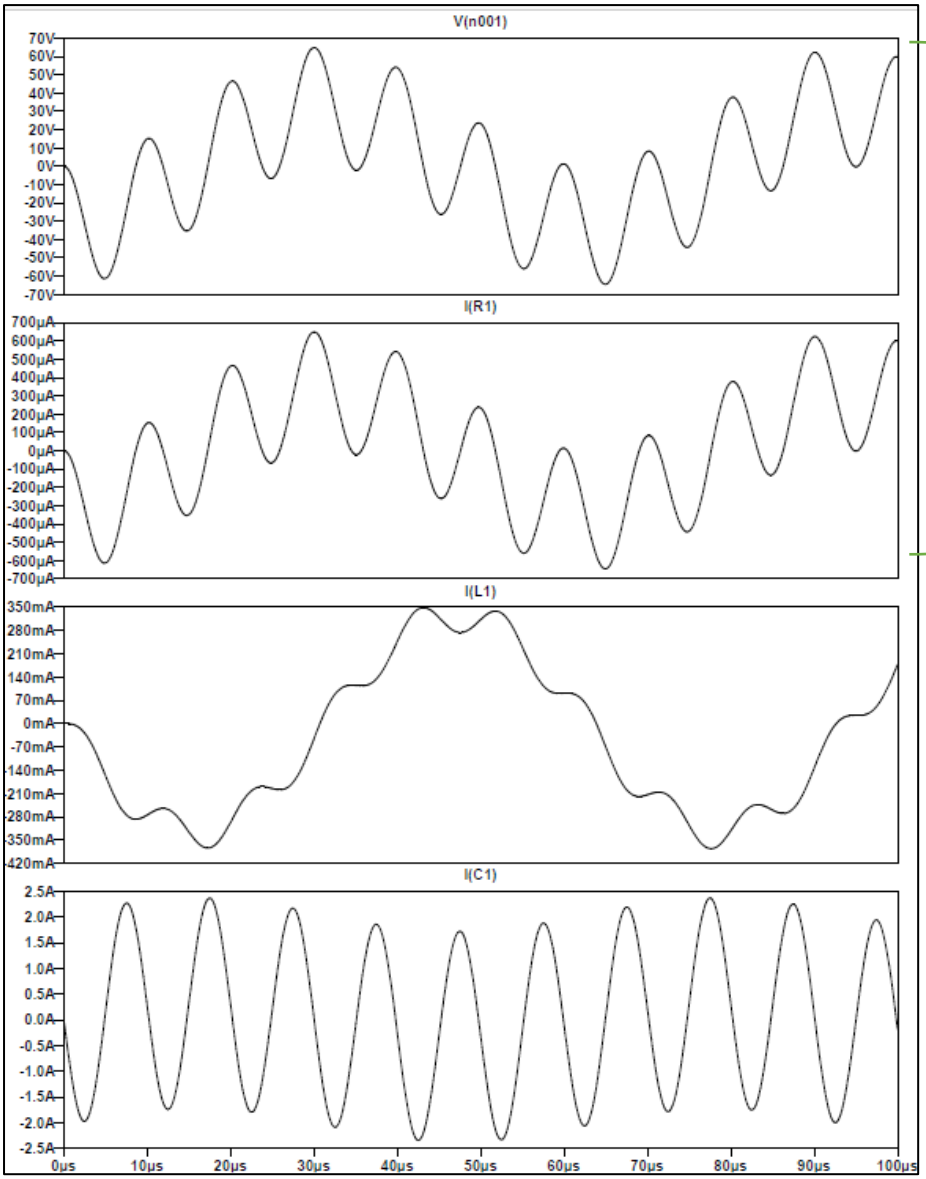


The magnitude at phase=-90 degree and resonance frequency both are matching with above calculated results.



Simulated results obtained for parallel resonance

Transient Analysis of Parallel RLC circuit:

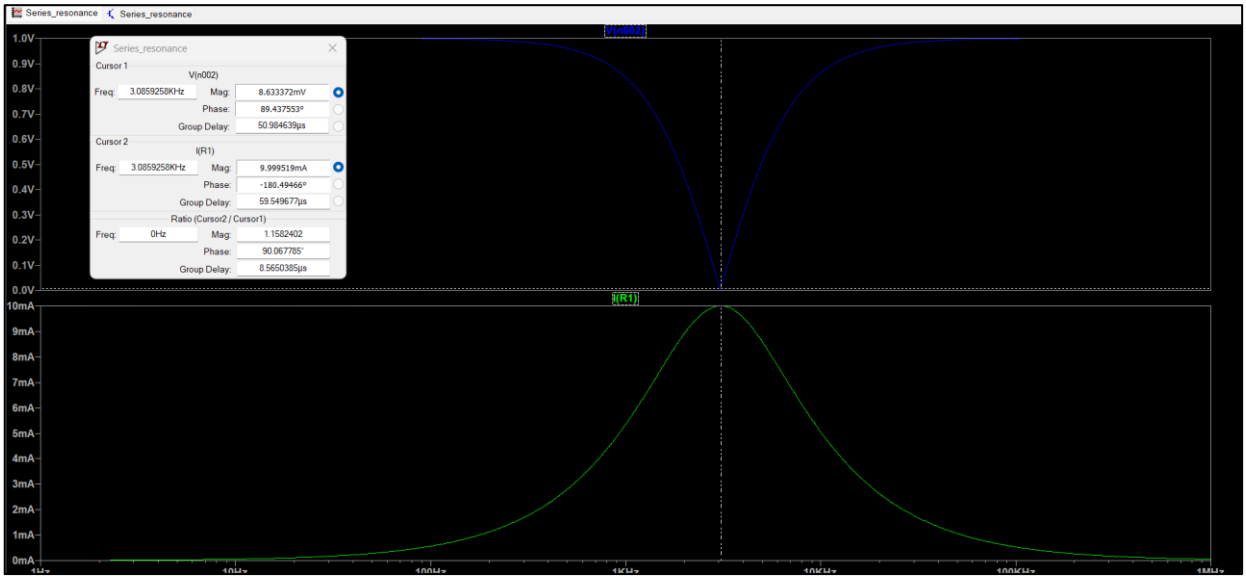


The current across resistance is in phase with Voltage across the circuit.

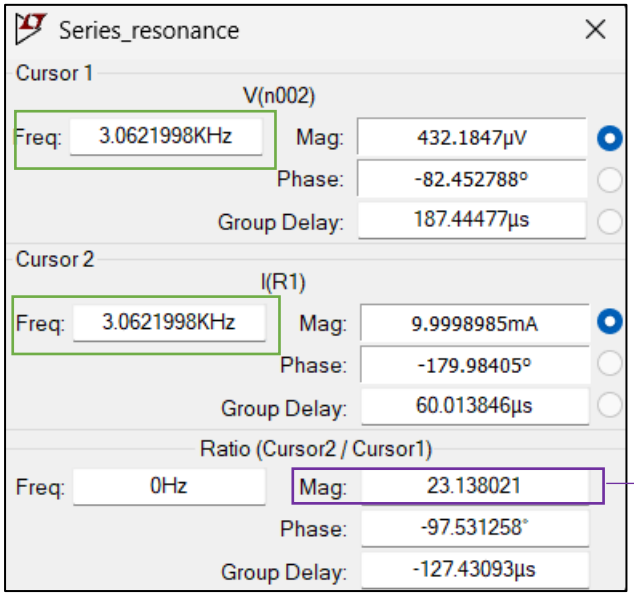
Transient Analysis of Parallel RLC circuit

Comparison of series RLC and Parallel RLC:

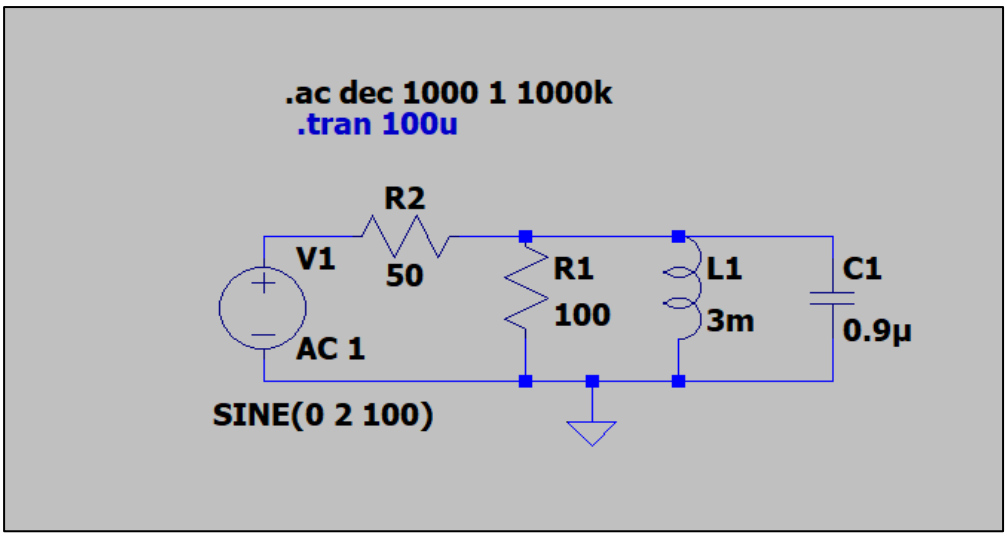
Parameter	Series	Parallel
Magnitude @ phase=-90 degree (in dB)	-4.77095	4.7721
Resonance Frequency	3.063KHZ	3.063KHZ
Quality factor	1.5019912	1.73205
Damping Ration (zeta)	0.2886	0.866

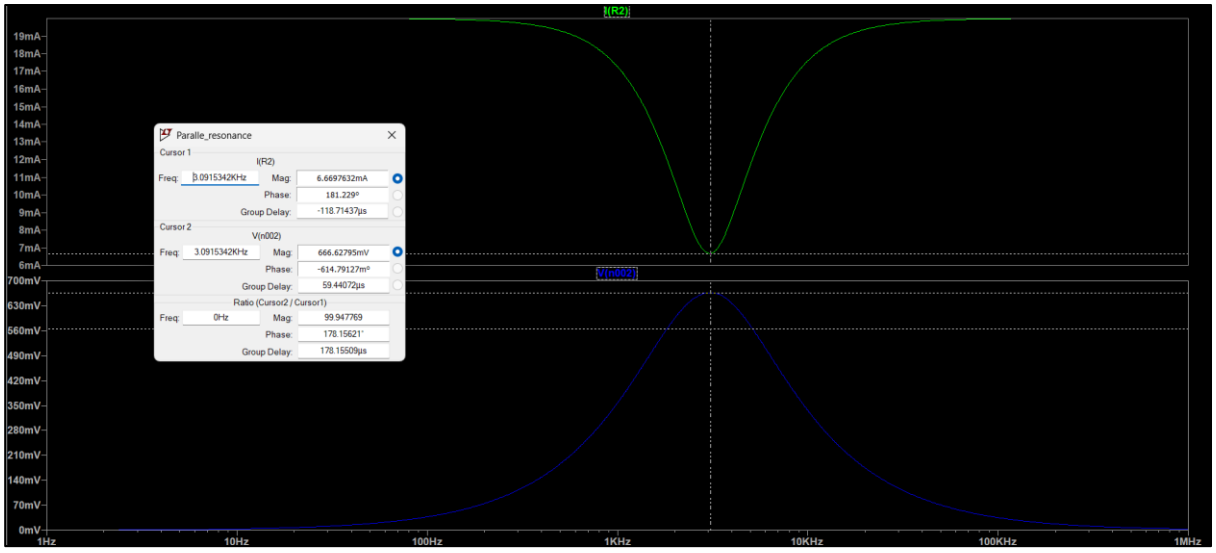


At resonance frequency current is maximum and impedance is minimum. [**This graph is series RLC Circuit**]



The Ratio of Vn002 and IR1 gives the impedance value and this is calculated at resonance frequency



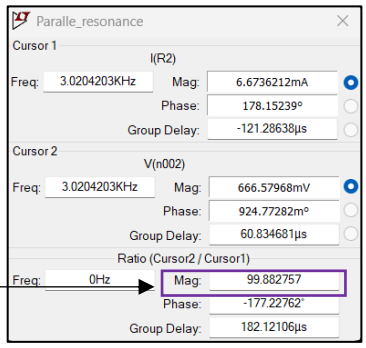


To obtain impedance value in **Parallel resonance** we have to replace current source with voltage source and have to add series resistance because voltage across current source is giving unwanted results. Also the LT spice is working **Modified nodal analysis**.

That's because LTspice uses the modified nodal analysis, where currents are analyzed, together with conductances & co. Mike Engelhardt, the author of LTspice, said repeatedly that voltage sources, which have (machine) zero internal resistance, pose problems in the matrix solver due to the inversion of this zero resistance. That's not to say voltage sources are prohibited, just that current sources will be superior in terms of convergence.

What is learned in school and how a solver in implemented in software, can be two very different things. And what you're quoting there is not part of the official help, but an addendum from a very knowledgeable member of the LTspice group. But you can read more straight from Mike, himself, from this link.

Source :https://electronics.stackexchange.com/questions/524042/why-do-



The Ratio of IR2 and V(n002) gives the admittance value and this is calculated at resonance frequency.

Conclusion: RLC Resonance & Transient Analysis

Summary of Results.

The study of both series and parallel RLC resonant circuits shows that the theoretical results and simulation results are in good agreement. The magnitude response and phase angle obtained from simulation match the expected values, especially at the point where the phase becomes -90°. The resonance frequency calculated theoretically is also very close to the value obtained from simulation. The transient analysis further confirms this, as the time response shows the expected oscillation and damping behavior of the circuit.

While performing simulations in LTspice, it was necessary to add a small series resistance to the inductor. This is because a real inductor always has some internal resistance. If we use an ideal inductor (with zero resistance), it can give unrealistic results and may also cause simulation convergence problems. Adding a small resistance makes the circuit more practical and stable.

In the case of the parallel RLC circuit, using only an ideal current source or voltage source is sometimes not enough to get proper results. This happens because the circuit may not have a proper reference path, leading to incorrect or undefined values. Therefore, a voltage source with a series element (or proper connection) is added to ensure correct measurement of voltages and currents.

Overall, including practical components and proper source connections helps in getting accurate and reliable results for RLC resonance circuits.

Values Considered during experiment:

Applicable to both series RLC and Parallel RLC

For Transient Analysis:

Voltage Source = 1V AC {During Small signal Analysis}.

Current Source = 1A AC {During small signal analysis of Parallel RLC }

Resistance = 100ohm , **Inductance** = 3 millihenry and **Capacitance** = 0.9 microfarad.

Transient time period: 100 microsecond.

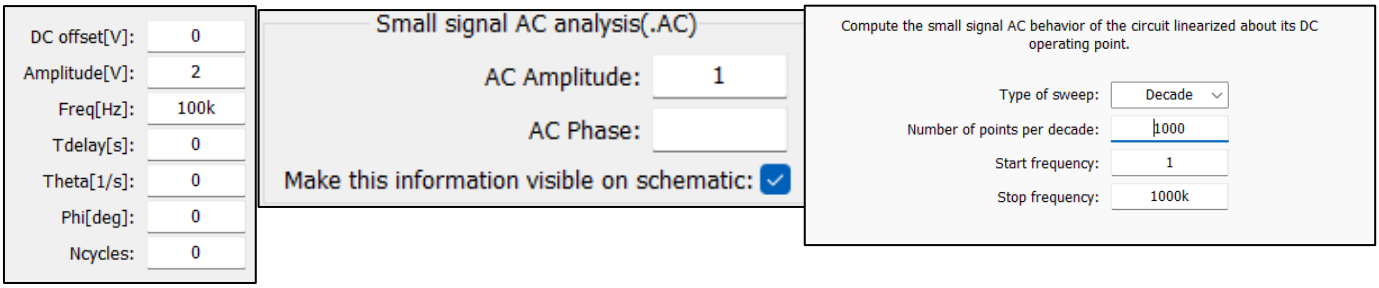
For AC Analysis:

Type of Sweep : Decade

Number of point per Decade = 1000

Start Frequency: 1Hz

Stop Frequency :1000KHz

Note: For finding minimum admittance or maximum impedance in parallel RLC circuit we replaced current source with voltage source and series resistance of:

Voltage Source: 1V **Series Resistance R2** = 50 ohm.